

Shannon–Rényi Control of the Arithmetic Power Geometry Closure Defect

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Abstract

This paper proves a Shannon–Rényi control theorem for the closure-defect functional of Arithmetic Power Geometry (APG). APG studies algebraic closure relations under exponent deformation by fixing positive base coordinates and allowing the exponent to vary as a deformation parameter. We first establish an exact bridge identity showing that the APG local closure defect at the Euclidean target is identical to a normalized power-sum defect associated with the probability vector determined by the squared base weights. This converts the APG defect into an information-theoretic object. Using Rényi entropy monotonicity, we prove two-sided bounds for the APG closure defect in terms of the maximum coordinate weight and Shannon entropy. We then derive explicit integrated-defect bounds for the APG global functional A_0 , including the quadratic estimate

$$A_0(a, b; n) \leq \frac{H(W)}{4} (n - 2)^2.$$

We also prove a Shannon–variance asymptotic law identifying the second-order coefficient of the local APG defect. The paper does not claim consequences for Fermat’s Last Theorem, the abc conjecture, the Szpiro conjecture, or the Birch and Swinnerton-Dyer conjecture; rather, it establishes foundational inequalities and asymptotics showing that APG deformation is quantitatively controlled by Shannon–Rényi information structure.

Keywords: Arithmetic Power Geometry; closure defect; exponent deformation; Shannon entropy; Rényi entropy; information geometry; power sums; Diophantine geometry; APG invariant; integrated defect.

1 Introduction

Arithmetic Power Geometry (APG) is a framework for studying algebraic closure relations under exponent deformation. Instead of fixing an exponent and varying coordinates, APG fixes positive base coordinates and studies how closure changes when the exponent moves away from the Euclidean baseline $n = 2$ [1, 2]. This viewpoint is motivated by the structural contrast between the Euclidean relation

$$a^2 + b^2 = c^2,$$

where closure is abundant, and higher-power regimes, where exact arithmetic closure becomes highly restricted [15, 16, 17, 23].

The early APG program introduced the Power-Deformation Space, Euclidean Target, local closure defect, entropy-governed asymptotics, information-geometric metrics, integrated defect functionals, scale-corrected invariants, arithmetic regularization, and conditional APG–Arakelov projection structures [1, 2, 3, 4, 5, 6, 7]. APG 0 then consolidated these ideas into a unified foundation while explicitly distinguishing proved internal results from conditional height-coupling and projection programs [8].

The present paper proves a clean theorem connecting APG directly to classical information theory. Specifically, we show that the APG local closure defect is exactly a power-sum defect of a probability distribution determined by the squared coordinate weights. This identity allows the defect to be bounded using Rényi entropy and Shannon entropy [21, 22]. The resulting theorem provides the explicit control law

$$1 - M^{(t-2)/2} \leq \delta_K(t) \leq 1 - \exp\left(-\frac{t-2}{2}H(W)\right) \leq \frac{t-2}{2}H(W).$$

The integrated APG defect consequently satisfies

$$A_0(a, b; n) \leq \frac{H(W)}{4}(n-2)^2.$$

The paper also proves a local Shannon–variance expansion for the APG defect and for its logarithmic defect strain. This strengthens the first-order entropy asymptotics of APG I by identifying the second-order information-theoretic curvature term. The result does not solve a major Diophantine conjecture, but it supplies a rigorous theorem that can be read, checked, and used independently of the larger APG program.

2 Background and Motivation

Information theory began with Shannon’s entropy as a measure of uncertainty in a probability distribution [22]. Rényi later introduced a one-parameter family of entropies that interpolates between several information measures and captures power-sum behavior [21]. These quantities now appear throughout probability theory, coding theory, statistical inference, information geometry, thermodynamics, and mathematical inequalities [9, 10, 12, 13, 20, 24].

APG naturally produces a probability vector from positive coordinates. Given $a, b > 0$, define

$$w_a = \frac{a^2}{a^2 + b^2}, \quad w_b = \frac{b^2}{a^2 + b^2}.$$

Then $W = (w_a, w_b)$ is a probability distribution. In APG, this vector measures the relative structural contribution of each coordinate at the Euclidean baseline. Earlier APG work observed that the Shannon entropy

$$H(W) = -(w_a \log w_a + w_b \log w_b)$$

governs the local deformation rate near $n = 2$ [1, 2]. The present paper strengthens that connection by proving a global entropy-control inequality for all $t \geq 2$.

The result uses only standard facts about power sums, entropy, and Rényi monotonicity. It therefore connects APG to established mathematical literature on inequalities, majorization, convexity, and information measures [11, 14, 18, 19].

3 APG Closure Defect and Normalized Weights

Let $a, b > 0$. The Euclidean Target associated with the pair (a, b) is

$$c_2 = \sqrt{a^2 + b^2}.$$

For a real exponent $t > 0$, the APG local closure defect at the Euclidean Target is

$$\delta_K(t) = \left| 1 - \frac{a^t + b^t}{(a^2 + b^2)^{t/2}} \right|.$$

This defect measures how much the Euclidean closure coordinate fails to remain closed when the exponent changes from 2 to t . At $t = 2$, one has $\delta_K(2) = 0$. For positive a and b , the defect is strictly positive for every $t > 2$.

Define

$$w_a = \frac{a^2}{a^2 + b^2}, \quad w_b = \frac{b^2}{a^2 + b^2}.$$

Then $w_a, w_b \in (0, 1)$ and $w_a + w_b = 1$. The corresponding Shannon entropy is

$$H(W) = -(w_a \log w_a + w_b \log w_b),$$

and the maximum coordinate weight is

$$M = \max(w_a, w_b).$$

The normalized APG defect is defined by

$$D(t) = 1 - (w_a^{t/2} + w_b^{t/2}).$$

The first theorem proves that $D(t)$ is not merely an auxiliary expression: it is exactly the APG local closure defect for $t \geq 2$.

4 Bridge Theorem

Theorem 4.1 (APG Normalization Bridge Identity). *Let $a, b > 0$, and define*

$$w_a = \frac{a^2}{a^2 + b^2}, \quad w_b = \frac{b^2}{a^2 + b^2}.$$

For every $t \geq 2$, the APG local closure defect satisfies

$$\delta_K(t) = D(t),$$

where

$$D(t) = 1 - (w_a^{t/2} + w_b^{t/2}).$$

Proof. By definition,

$$w_a^{t/2} = \left(\frac{a^2}{a^2 + b^2} \right)^{t/2} = \frac{a^t}{(a^2 + b^2)^{t/2}}.$$

Similarly,

$$w_b^{t/2} = \left(\frac{b^2}{a^2 + b^2} \right)^{t/2} = \frac{b^t}{(a^2 + b^2)^{t/2}}.$$

Adding gives

$$w_a^{t/2} + w_b^{t/2} = \frac{a^t + b^t}{(a^2 + b^2)^{t/2}}.$$

Therefore,

$$\delta_K(t) = \left| 1 - \frac{a^t + b^t}{(a^2 + b^2)^{t/2}} \right| = \left| 1 - (w_a^{t/2} + w_b^{t/2}) \right|.$$

For $t \geq 2$, we have $t/2 \geq 1$. Since $0 < w_a, w_b < 1$, it follows that

$$w_a^{t/2} \leq w_a, \quad w_b^{t/2} \leq w_b.$$

Hence

$$w_a^{t/2} + w_b^{t/2} \leq w_a + w_b = 1.$$

Thus the absolute value drops, and

$$\delta_K(t) = 1 - (w_a^{t/2} + w_b^{t/2}) = D(t).$$

This proves the bridge identity. □

5 Shannon–Rényi Control Theorem

Theorem 5.1 (Shannon–Rényi Bound for the APG Closure Defect). *Let $a, b > 0$, let $W = (w_a, w_b)$, and let*

$$H(W) = -(w_a \log w_a + w_b \log w_b)$$

be the Shannon entropy of W . Let

$$M = \max(w_a, w_b).$$

Then for every $t \geq 2$,

$$1 - M^{(t-2)/2} \leq D(t) \leq 1 - \exp\left(-\frac{t-2}{2}H(W)\right).$$

Consequently,

$$D(t) \leq \frac{t-2}{2}H(W).$$

Since $D(t) = \delta_K(t)$, the same bounds hold for the APG local closure defect.

Proof. Set

$$q = \frac{t}{2} \geq 1.$$

Then

$$D(t) = 1 - (w_a^q + w_b^q) = 1 - \sum_i w_i^q.$$

First, since $w_i \leq M$, we have

$$w_i^q = w_i w_i^{q-1} \leq w_i M^{q-1}.$$

Summing over i gives

$$\sum_i w_i^q \leq M^{q-1} \sum_i w_i = M^{q-1}.$$

Therefore,

$$D(t) = 1 - \sum_i w_i^q \geq 1 - M^{q-1}.$$

Since $q = t/2$,

$$q - 1 = \frac{t - 2}{2}.$$

Thus

$$D(t) \geq 1 - M^{(t-2)/2}.$$

For the upper bound, define the Rényi entropy of order $q \neq 1$ by

$$H_q(W) = \frac{1}{1-q} \log \left(\sum_i w_i^q \right).$$

For $q \geq 1$, Rényi entropy is bounded above by Shannon entropy:

$$H_q(W) \leq H(W).$$

Because $1 - q \leq 0$, multiplying by $1 - q$ reverses the inequality:

$$\log \left(\sum_i w_i^q \right) \geq (1 - q) H(W).$$

Exponentiating gives

$$\sum_i w_i^q \geq \exp((1 - q)H(W)) = \exp(-(q - 1)H(W)).$$

Therefore,

$$D(t) = 1 - \sum_i w_i^q \leq 1 - \exp(-(q - 1)H(W)).$$

Substituting $q = t/2$ yields

$$D(t) \leq 1 - \exp\left(-\frac{t-2}{2}H(W)\right).$$

Finally, for $x \geq 0$, the elementary inequality $1 - e^{-x} \leq x$ gives

$$D(t) \leq \frac{t-2}{2} H(W).$$

This completes the proof. □

6 Integrated APG Defect Bound

The integrated APG closure defect is defined by

$$A_0(a, b; n) = \int_2^n D(t) dt.$$

This functional was introduced in APG as a global measure of accumulated exponent-deformation defect [3]. The Shannon–Rényi theorem gives a direct bound for A_0 .

Theorem 6.1 (Integrated Shannon–Rényi Bound). *For $a, b > 0$ and $n \geq 2$,*

$$A_0(a, b; n) \leq (n-2) - \frac{2}{H(W)} \left(1 - \exp \left(-\frac{(n-2)H(W)}{2} \right) \right),$$

with the limiting interpretation that the right-hand side tends to 0 when $H(W) \rightarrow 0$. Moreover,

$$A_0(a, b; n) \leq \frac{H(W)}{4} (n-2)^2.$$

Proof. By Theorem 5.1,

$$D(t) \leq 1 - \exp \left(-\frac{t-2}{2} H(W) \right).$$

Therefore,

$$A_0(a, b; n) = \int_2^n D(t) dt \leq \int_2^n \left[1 - \exp \left(-\frac{t-2}{2} H(W) \right) \right] dt.$$

Let $u = t - 2$. Then $u \in [0, n-2]$, and the integral becomes

$$\int_0^{n-2} \left[1 - \exp \left(-\frac{H(W)}{2} u \right) \right] du.$$

Evaluating,

$$\int_0^{n-2} 1 du = n-2,$$

and

$$\int_0^{n-2} \exp \left(-\frac{H(W)}{2} u \right) du = \frac{2}{H(W)} \left(1 - \exp \left(-\frac{(n-2)H(W)}{2} \right) \right).$$

Hence

$$A_0(a, b; n) \leq (n-2) - \frac{2}{H(W)} \left(1 - \exp \left(-\frac{(n-2)H(W)}{2} \right) \right).$$

The simpler estimate follows from

$$D(t) \leq \frac{t-2}{2} H(W).$$

Therefore,

$$A_0(a, b; n) \leq \int_2^n \frac{t-2}{2} H(W) dt = \frac{H(W)}{2} \int_2^n (t-2) dt.$$

Since

$$\int_2^n (t-2) dt = \frac{(n-2)^2}{2},$$

we obtain

$$A_0(a, b; n) \leq \frac{H(W)}{4} (n-2)^2.$$

This proves the theorem. □

7 Shannon–Variance Asymptotics

The preceding bounds give global Shannon–Rényi control of the APG closure defect. We now record a sharper local expansion at the Euclidean target. This result identifies not only the first-order entropy coefficient but also the second-order information-theoretic curvature term.

Theorem 7.1 (Shannon–Variance Asymptotic Law for the APG Closure Defect). *Let $W = (w_i)$ be a finite probability vector with $w_i > 0$ and $\sum_i w_i = 1$. Define*

$$D(t) = 1 - \sum_i w_i^{t/2},$$

$$H(W) = - \sum_i w_i \log w_i,$$

and

$$V(W) = \sum_i w_i (\log w_i)^2 - H(W)^2.$$

Then, as $t \rightarrow 2$,

$$D(t) = \frac{H(W)}{2} (t-2) - \frac{V(W) + H(W)^2}{8} (t-2)^2 + O((t-2)^3).$$

Moreover, if the APG logarithmic defect strain is defined by

$$L(t) = -\log \left(\sum_i w_i^{t/2} \right),$$

then

$$L(t) = \frac{H(W)}{2} (t-2) - \frac{V(W)}{8} (t-2)^2 + O((t-2)^3).$$

Proof. Set $q = t/2$ and write $q = 1 + \varepsilon$, where

$$\varepsilon = \frac{t-2}{2}.$$

Then

$$\sum_i w_i^q = \sum_i w_i^{1+\varepsilon} = \sum_i w_i \exp(\varepsilon \log w_i).$$

Using the Taylor expansion of the exponential function,

$$\sum_i w_i^q = 1 + \varepsilon \sum_i w_i \log w_i + \frac{\varepsilon^2}{2} \sum_i w_i (\log w_i)^2 + O(\varepsilon^3).$$

Since $\sum_i w_i \log w_i = -H(W)$, this becomes

$$\sum_i w_i^q = 1 - \varepsilon H(W) + \frac{\varepsilon^2}{2} \sum_i w_i (\log w_i)^2 + O(\varepsilon^3).$$

Therefore

$$D(t) = 1 - \sum_i w_i^q = \varepsilon H(W) - \frac{\varepsilon^2}{2} \sum_i w_i (\log w_i)^2 + O(\varepsilon^3).$$

By definition of the entropy variance,

$$\sum_i w_i (\log w_i)^2 = V(W) + H(W)^2.$$

Substituting this identity and then $\varepsilon = (t - 2)/2$ gives

$$D(t) = \frac{H(W)}{2}(t - 2) - \frac{V(W) + H(W)^2}{8}(t - 2)^2 + O((t - 2)^3).$$

For the logarithmic defect strain, write

$$S(\varepsilon) = \sum_i w_i^{1+\varepsilon}.$$

Then $S(0) = 1$,

$$S'(\varepsilon) = \sum_i w_i^{1+\varepsilon} \log w_i, \quad S''(\varepsilon) = \sum_i w_i^{1+\varepsilon} (\log w_i)^2.$$

Hence

$$S'(0) = \sum_i w_i \log w_i = -H(W),$$

and

$$S''(0) = \sum_i w_i (\log w_i)^2.$$

For $G(\varepsilon) = \log S(\varepsilon)$, we have

$$G'(\varepsilon) = \frac{S'(\varepsilon)}{S(\varepsilon)},$$

and

$$G''(\varepsilon) = \frac{S''(\varepsilon)S(\varepsilon) - (S'(\varepsilon))^2}{S(\varepsilon)^2}.$$

At $\varepsilon = 0$, this gives

$$G'(0) = -H(W),$$

and

$$G''(0) = \sum_i w_i (\log w_i)^2 - H(W)^2 = V(W).$$

Taylor's theorem therefore yields

$$\log S(\varepsilon) = -\varepsilon H(W) + \frac{\varepsilon^2}{2} V(W) + O(\varepsilon^3).$$

Thus

$$L(t) = -\log S(\varepsilon) = \varepsilon H(W) - \frac{\varepsilon^2}{2} V(W) + O(\varepsilon^3).$$

Substituting $\varepsilon = (t - 2)/2$ proves

$$L(t) = \frac{H(W)}{2}(t - 2) - \frac{V(W)}{8}(t - 2)^2 + O((t - 2)^3).$$

This completes the proof. □

Corollary 7.2 (Second-Order Entropy Control). *Because $V(W) \geq 0$, the logarithmic defect strain has the second-order expansion*

$$L(t) = \frac{H(W)}{2}(t - 2) - \frac{V(W)}{8}(t - 2)^2 + O((t - 2)^3)$$

near $t = 2$. Hence Shannon entropy is the first-order generator of APG deformation, while entropy variance gives the second-order correction term.

Corollary 7.3 (Balanced Configuration). *If $w_i = 1/m$ for all i , then $V(W) = 0$ and $H(W) = \log m$. Therefore*

$$L(t) = \frac{\log m}{2}(t - 2) + O((t - 2)^3).$$

In the two-coordinate balanced case, this recovers the sharp exponential Shannon–Rényi bound appearing in Example 8.2.

Corollary 7.4 (Entropy-Collapse Regime). *If W approaches a degenerate distribution, then $H(W) \rightarrow 0$ and $V(W) \rightarrow 0$. Consequently both $D(t)$ and $L(t)$ vanish to second order near the Euclidean target, matching the APG entropy-collapse boundary.*

8 Numerical Examples

Example 8.1 (Classical (3, 4) configuration). *Let $a = 3$ and $b = 4$. Then*

$$w_a = \frac{9}{25} = 0.36, \quad w_b = \frac{16}{25} = 0.64.$$

The Shannon entropy is

$$H(W) = -(0.36 \log 0.36 + 0.64 \log 0.64) \approx 0.6534.$$

At $t = 3$,

$$D(3) = 1 - (0.36^{3/2} + 0.64^{3/2}) \approx 0.2723.$$

The Shannon–Rényi upper bound gives

$$1 - \exp\left(-\frac{1}{2}H(W)\right) \approx 1 - e^{-0.3267} \approx 0.2788.$$

Thus

$$D(3) \approx 0.2723 \leq 0.2788.$$

The bound is close, showing that Shannon entropy provides an effective quantitative control of the APG defect.

Example 8.2 (Balanced configuration). *Let $a = b$. Then*

$$w_a = w_b = \frac{1}{2}, \quad H(W) = \log 2.$$

For $t \geq 2$,

$$D(t) = 1 - 2\left(\frac{1}{2}\right)^{t/2} = 1 - 2^{1-t/2}.$$

The upper entropy bound becomes

$$D(t) \leq 1 - \exp\left(-\frac{t-2}{2}\log 2\right) = 1 - 2^{-(t-2)/2} = 1 - 2^{1-t/2}.$$

Thus equality holds in the balanced case for the exponential Shannon–Rényi upper bound. This shows that the exponential upper bound is sharp, while the simpler linear estimate is sharp only infinitesimally at $t = 2$.

Example 8.3 (Extreme skew limit). *Suppose $a/b \rightarrow 0$. Then*

$$w_a \rightarrow 0, \quad w_b \rightarrow 1, \quad H(W) \rightarrow 0.$$

The theorem gives

$$D(t) \leq \frac{t-2}{2}H(W) \rightarrow 0.$$

Thus the APG defect collapses in the extreme skew limit. This agrees with the entropy-collapse phenomenon identified in the APG framework [4, 5].

9 Discussion

The main result shows that the APG closure defect is not an arbitrary deformation quantity. It is exactly a power-sum defect of a canonical probability distribution. This places the APG local defect within the established mathematical theory of Shannon entropy, Rényi entropy, convexity, and majorization.

The theorem has three important consequences. First, it proves that APG local deformation is entropy-controlled. The bound

$$D(t) \leq \frac{t-2}{2}H(W)$$

shows that coordinate entropy limits the maximum possible defect growth away from the Euclidean baseline.

Second, the integrated defect satisfies

$$A_0(a, b; n) \leq \frac{H(W)}{4}(n-2)^2.$$

This gives a global deformation-energy estimate in terms of Shannon entropy alone.

Third, the exponential Shannon–Rényi upper bound is sharp in the balanced case $a = b$. Thus the theorem is not merely qualitative; it identifies an extremal entropy configuration where the exponential estimate becomes exact.

The Shannon–variance asymptotic law further shows that the first-order local APG deformation rate is controlled by Shannon entropy, while the second-order correction is governed by entropy variance. This supplies a more refined information-theoretic description of the APG defect near the Euclidean target.

This result is modest in scope but structurally important. It does not prove a Diophantine obstruction theorem. It does not solve Fermat’s Last Theorem. It does not imply the abc or Szpiro conjectures. Its significance is that it supplies a rigorous external mathematical control law for an APG invariant using classical information theory.

10 Open Problems

Several questions naturally follow.

1. Can the Shannon–Rényi control theorem be extended from two-coordinate APG systems to m -coordinate exponent-deformation systems?
2. Can sharper lower bounds be obtained in terms of Shannon entropy alone, without using the maximum coordinate weight M ?
3. Does the integrated bound for A_0 admit an exact extremal characterization for fixed entropy?
4. Can the entropy-controlled APG defect be connected to classical height functions in a nontrivial unconditional way?
5. Can the Shannon–Rényi bound be used to derive stability estimates for APG metric tensors?
6. Can APG defect control be generalized to weighted L^p geometry, Orlicz spaces, or information-geometric divergence structures?
7. Can the entropy-collapse boundary be used to classify degenerate APG deformation regimes?
8. Is there a direct relation between the APG defect hierarchy and majorization order?

These problems suggest that APG may serve as a bridge between exponent-deformation geometry and entropy-based inequality theory.

11 Conclusion

This paper proved the Shannon–Rényi control theorem for the APG closure defect. The central bridge identity shows that the APG local closure defect at the Euclidean Target is exactly the normalized power-sum defect of the squared-coordinate probability distribution. Rényi entropy monotonicity then yields two-sided estimates for the defect, and Shannon entropy provides a simple global upper control. Integrating these bounds gives explicit estimates for the integrated APG defect A_0 , including

$$A_0(a, b; n) \leq \frac{H(W)}{4}(n-2)^2.$$

The paper also proves the Shannon–variance asymptotic law

$$D(t) = \frac{H(W)}{2}(t-2) - \frac{V(W) + H(W)^2}{8}(t-2)^2 + O((t-2)^3),$$

and the logarithmic defect-strain expansion

$$L(t) = \frac{H(W)}{2}(t-2) - \frac{V(W)}{8}(t-2)^2 + O((t-2)^3).$$

The result establishes a rigorous mathematical link between APG exponent deformation and classical information theory. The theorem is externally checkable, has a sharp exponential bound in the balanced case, and is compatible with the entropy-collapse behavior identified in earlier APG work. It should therefore be understood as a foundational APG control theorem rather than as a claim about major Diophantine conjectures.

Author’s Note

The conceptual origins of Arithmetic Power Geometry (APG) trace back to mathematical notes and exploratory investigations begun by the author in the early 1990s. The APG manuscript series represents a recent effort to consolidate, formalize, and communicate these ideas within a unified mathematical framework. The present paper should therefore be viewed as one component of a broader long-term research program whose development has occurred over many years.

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